

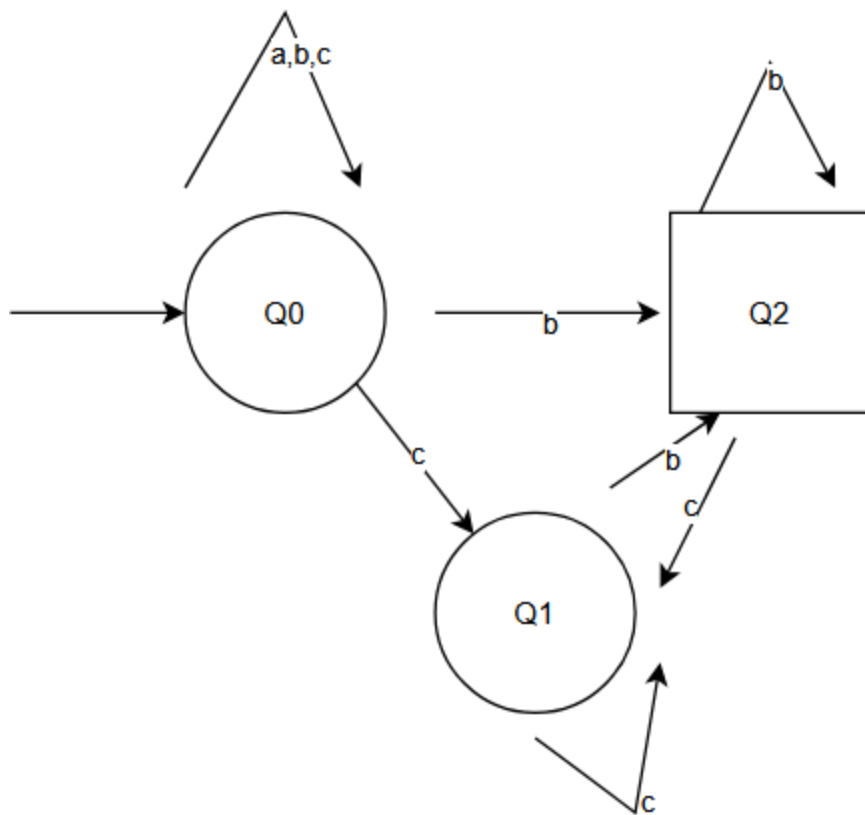
CPS I — Exercise Sheet 3

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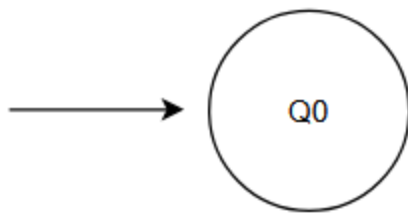
1. Intersection of ω -regular languages

(a)

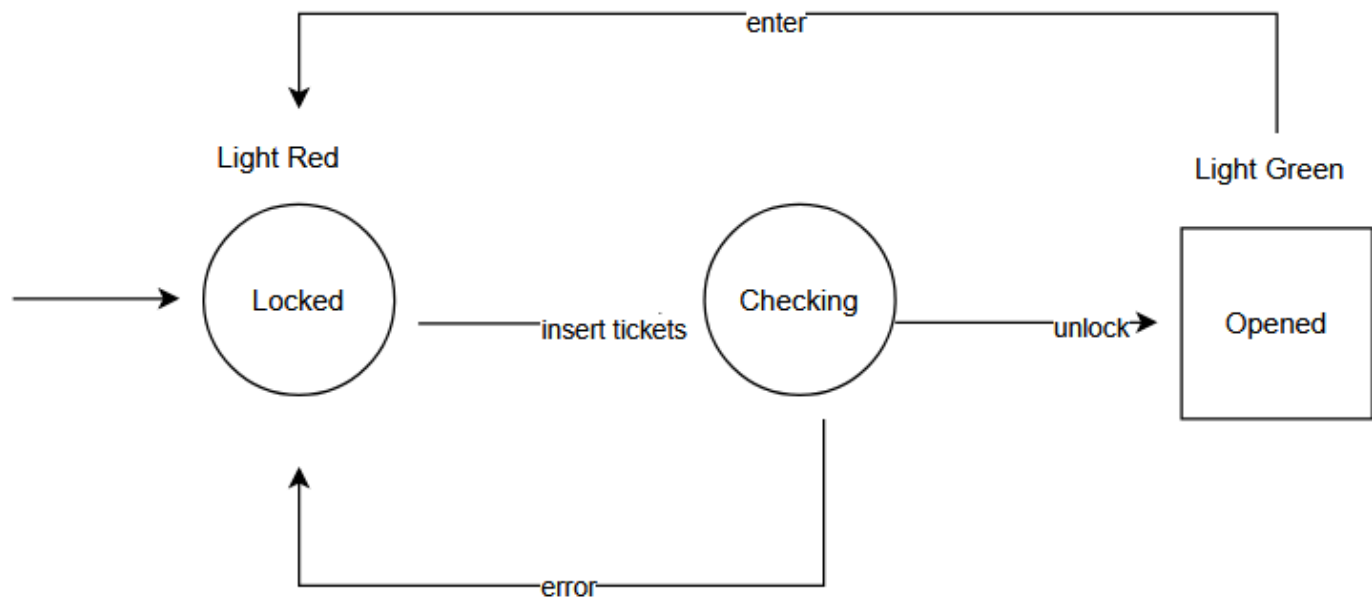


(b)

Empty ω Language

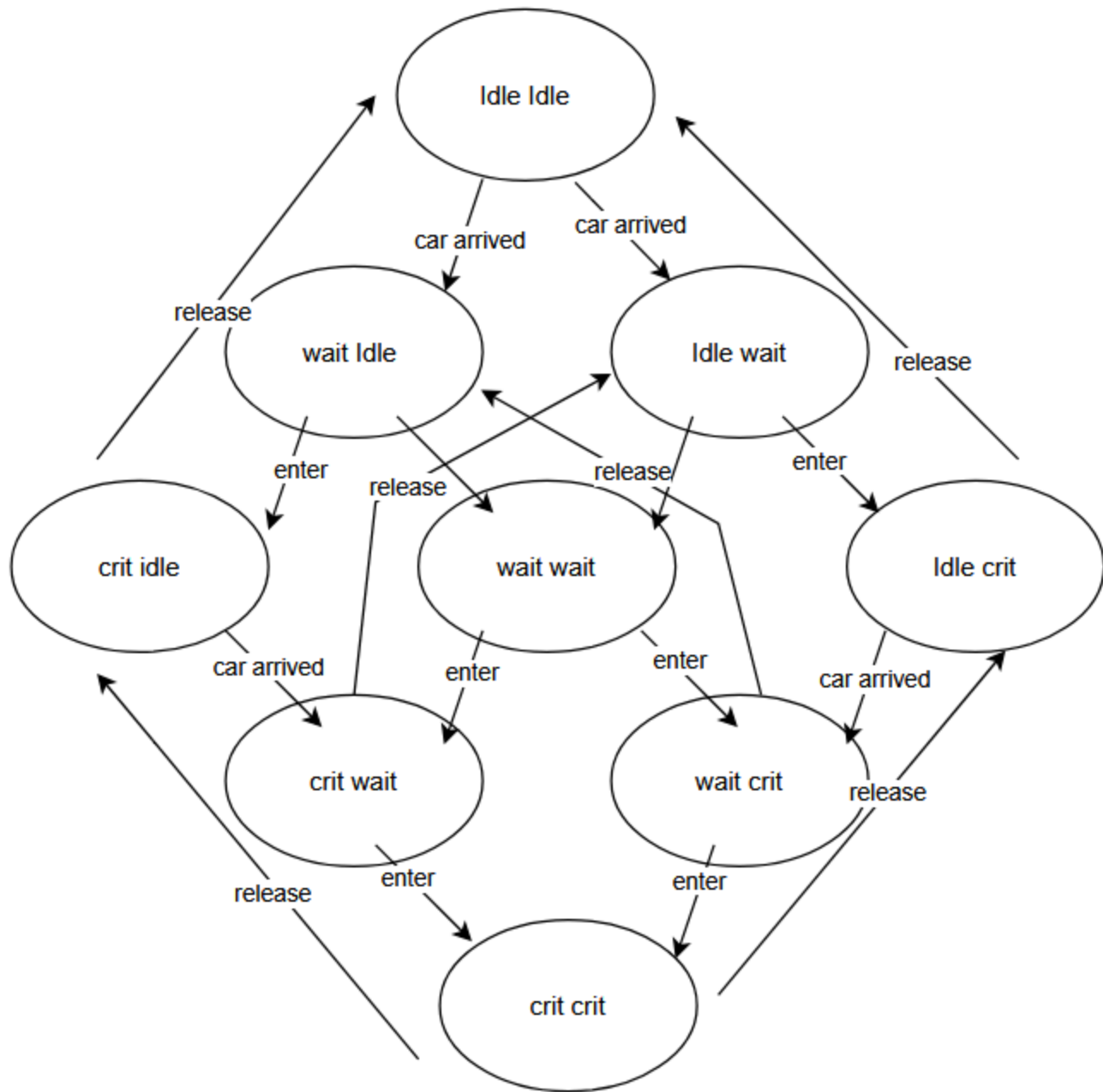


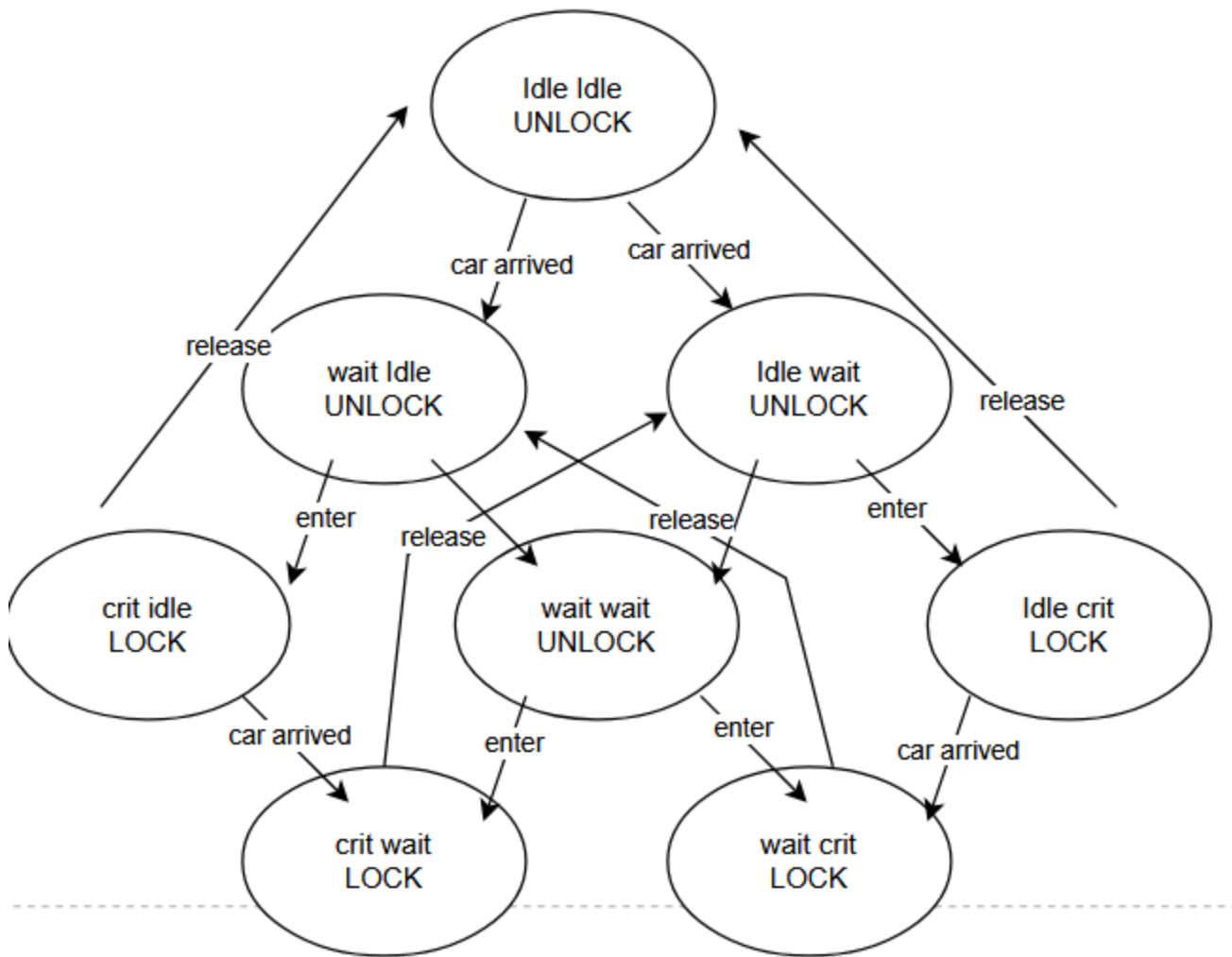
2. Transition Systems



This models an automated ticket gate

3. Crossroads Traffic Lights





- Yes, safe. The atomic propositions `green1` and `green2` correspond to `crit_1` and `crit_2`, respectively. Once one direction acquires the lock, the other direction cannot execute unlock until the first one releases it. Therefore, there is no reachable state where both `crit_1` and `crit_2` hold simultaneously.
- A common expectation is liveness / responsiveness / no starvation, i.e., if there is a car waiting in a direction (state `wait_i`), it will eventually be allowed to proceed (state `crit_i`). This property may not hold in the current model because the arbiter only enforces mutual exclusion but does not implement fair scheduling.